

09. Support Vector Machines (SVM)

Algorithms implemented: Support Vector Classification (SVC) with linear/poly/RBF kernels and Support Vector Regression (SVR) with an epsilon-insensitive tube.

Datasets: Synthetic two-moons (2D), Breast Cancer (30 features), and a 1D sinusoidal benchmark.

```
# ---- Core numerical and plotting libraries ----
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# ---- Datasets + model selection ----
from sklearn.datasets import load_breast_cancer, make_moons
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.preprocessing import StandardScaler
# ---- SVM models: classification and regression ----
from sklearn.svm import SVC, SVR
# ---- Evaluation metrics ----
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix, r2_score,
root_mean_squared_error

# Styling setup (consistent look across all notebooks)
plt.style.use('seaborn-v0_8-whitegrid' if 'seaborn-v0_8-whitegrid' in plt.style.available else 'default')
plt.rcParams['font.sans-serif'] = 'DejaVu Sans'
plt.rcParams['figure.figsize'] = (10, 6)
plt.rcParams['figure.dpi'] = 110

np.random.seed(42)
print("SVM modules loaded successfully!")
```

SVM modules loaded successfully!

1. Support Vector Classification (SVC)

Step-by-Step Algorithm (Support Vector Classification)

1. **Formulate** the soft-margin problem: minimize $\frac{1}{2}\|w\|^2 + C \sum \xi_i$ subject to $y_i(w^T\phi(x_i) + b) \geq 1 - \xi_i$, $\xi_i \geq 0$.
2. **Solve** the dual quadratic program with Lagrange multipliers α_i (convex $\rightarrow$ global optimum).
3. **Identify support vectors:** training points with $\alpha_i > 0$; they lie on or inside the margin.
4. **Kernel trick:** replace inner products with $K(x_i, x_j)$ (linear, polynomial, RBF, ...) to model non-linear boundaries.
5. **Predict:** class = $\text{sign}(\sum \alpha_i y_i K(x_i, x) + b)$.
6. **Tune** C and γ by cross-validation to balance margin width and violations.

```
# ---- 2D synthetic data for visualizing margins and kernels ----
# Two interleaving half-moons cannot be separated by a straight line.
X_2d, y_2d = make_moons(n_samples=250, noise=0.20, random_state=42)
scaler_2d = StandardScaler()
X_2d_scaled = scaler_2d.fit_transform(X_2d)

kernels = ['linear', 'poly', 'rbf']
fig, axes = plt.subplots(1, 3, figsize=(18, 5))

# Coordinate grid used to draw the decision boundary as a contour
x_min, x_max = X_2d_scaled[:, 0].min() - 0.5, X_2d_scaled[:, 0].max() + 0.5
y_min, y_max = X_2d_scaled[:, 1].min() - 0.5, X_2d_scaled[:, 1].max() + 0.5
xx, yy = np.meshgrid(np.linspace(x_min, x_max, 200), np.linspace(y_min, y_max, 200))

for ax, k_name in zip(axes, kernels):
    # Train one SVC per kernel on the same data
    clf = SVC(kernel=k_name, C=1.0, gamma='scale', degree=3, random_state=42)
    clf.fit(X_2d_scaled, y_2d)

    # Signed distance to the hyperplane for every grid point
    Z = clf.decision_function(np.c_[xx.ravel(), yy.ravel()]).reshape(xx.shape)

    # Filled contour = decision regions; solid line = boundary (Z=0); dashed = margins (Z=±1)
```

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ax.contourf(xx, yy, Z, levels=np.linspace(Z.min(), Z.max(), 20), cmap='coolwarm', alpha=0.3)
ax.contour(xx, yy, Z, levels=[-1.0, 0.0, 1.0], linestyles=['--', '-', '--'], colors=['k', 'k', 'k'],
linewidths=[1.5, 2.5, 1.5])

# Scatter the data points (colored by class)
ax.scatter(X_2d_scaled[:, 0], X_2d_scaled[:, 1], c=y_2d, cmap='coolwarm', edgecolors='k', s=40, alpha=0.85)

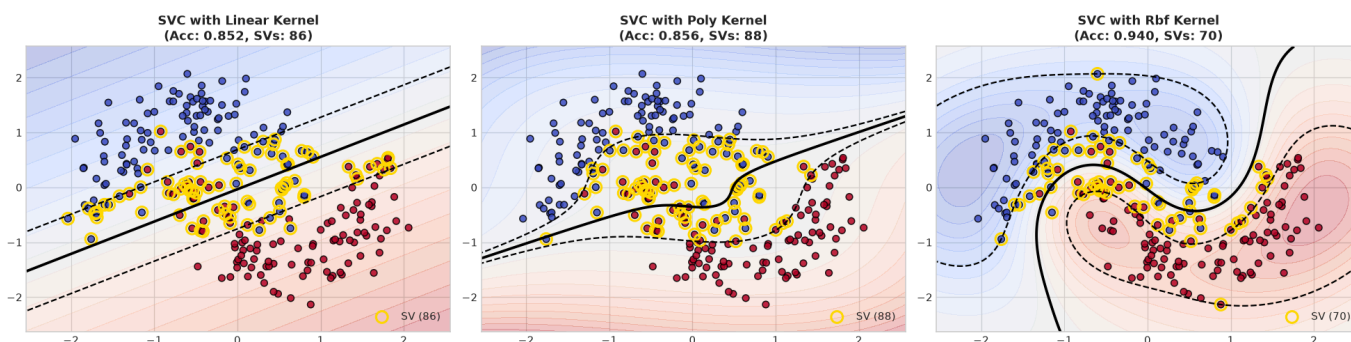
# Highlight the support vectors (the points that define the margin)
sv = clf.support_vectors_
ax.scatter(sv[:, 0], sv[:, 1], s=120, facecolors='none', edgecolors='gold', linewidths=2.0, label=f'SV
({len(sv)})')

ax.set_title(f"SVC with {k_name.capitalize()} Kernel\n(Acc: {clf.score(X_2d_scaled, y_2d):.3f}, SVs:
{len(sv)}", fontweight='bold')
ax.legend(loc='lower right')

plt.suptitle("SVC Decision Boundaries, Margin Tubes, and Support Vectors", fontsize=15, fontweight='bold')
plt.tight_layout()
plt.show()

```

SVC Decision Boundaries, Margin Tubes, and Support Vectors



```

# ---- SVC on a real dataset: Breast Cancer (RBF kernel) ----
cancer = load_breast_cancer()
X_train_c, X_test_c, y_train_c, y_test_c = train_test_split(
    cancer.data, cancer.target, test_size=0.25, random_state=42, stratify=cancer.target
)
scaler_c = StandardScaler()
X_train_c_s = scaler_c.fit_transform(X_train_c)
X_test_c_s = scaler_c.transform(X_test_c)

svc_rbf = SVC(kernel='rbf', C=1.0, random_state=42)
svc_rbf.fit(X_train_c_s, y_train_c)
print(f"SVC (RBF) test accuracy on Breast Cancer: {svc_rbf.score(X_test_c_s, y_test_c):.4f}")
print(f"Support vectors used: {len(svc_rbf.support_)} of {len(X_train_c)} training samples")

```

SVC (RBF) test accuracy on Breast Cancer: 0.9790
Support vectors used: 96 of 426 training samples

2. Support Vector Regression (SVR)

Step-by-Step Algorithm (Support Vector Regression)

1. Define an ϵ -insensitive tube: residuals with $|y_i - f(x_i)| \leq \epsilon$ incur zero loss.
2. Minimize $\frac{1}{2}\|w\|^2 + C \sum (\xi_i + \xi_i^*)$ subject to the tube constraints.
3. Solve the dual; only points on/outside the tube get $\alpha_i > 0$ and become support vectors.
4. **Predict:** $f(x) = \sum \alpha_i K(x_i, x) + b$ (kernel regression).
5. Visualize the fitted curve, the $\pm \epsilon$ tube, and the support vectors.

```

# ---- 1D non-linear regression demonstration ----
np.random.seed(42)
# Synthetic target: sine + linear trend + Gaussian noise
X_svr = np.sort(np.random.uniform(-3, 3, 120)).reshape(-1, 1)
y_svr = np.sin(X_svr).ravel() + 0.15 * X_svr.ravel() + np.random.normal(0, 0.12, 120)

# epsilon defines the width of the "no penalty" tube around the prediction
epsilon_val = 0.18
svr_model = SVR(kernel='rbf', C=20.0, epsilon=epsilon_val, gamma=0.5)
svr_model.fit(X_svr, y_svr)

# Dense grid for drawing a smooth prediction curve
X_grid = np.linspace(-3.2, 3.2, 300).reshape(-1, 1)
y_grid_pred = svr_model.predict(X_grid)

```

```

# Support vectors = training points that lie on/outside the epsilon tube
sv_indices = svr_model.support_
X_sv = X_svr[sv_indices]
y_sv = y_svr[sv_indices]

plt.figure(figsize=(11, 6))
# Training samples
plt.scatter(X_svr, y_svr, color='navy', s=35, alpha=0.75, label='Training Samples')

# Regression surface
plt.plot(X_grid, y_grid_pred, color='crimson', lw=2.5, label='SVR Prediction  $f(x)$ ')

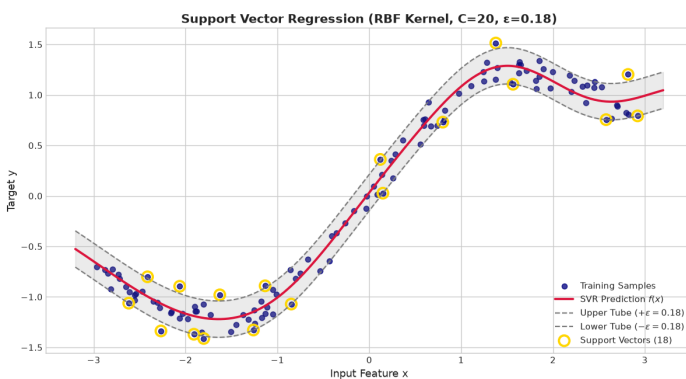
# Epsilon-insensitive tube boundaries + shaded tube
plt.plot(X_grid, y_grid_pred + epsilon_val, color='gray', linestyle='--', lw=1.5, label=rf'Upper Tube ($+\epsilon=\{epsilon\_val\}$)')
plt.plot(X_grid, y_grid_pred - epsilon_val, color='gray', linestyle='--', lw=1.5, label=rf'Lower Tube ($-\epsilon=\{epsilon\_val\}$)')
plt.fill_between(X_grid.ravel(), y_grid_pred - epsilon_val, y_grid_pred + epsilon_val, color='gray', alpha=0.15)

# Support vectors are highlighted - only these points influence the fitted curve
plt.scatter(X_sv, y_sv, s=140, facecolors='none', edgecolors='gold', linewidths=2.5, label=f'Support Vectors ({len(X_sv)})')

plt.title(f"Support Vector Regression (RBF Kernel, C=20,  $\epsilon=\{epsilon\_val\}$ )", fontsize=14, fontweight='bold')
plt.xlabel("Input Feature x", fontsize=12)
plt.ylabel("Target y", fontsize=12)
plt.legend(loc='lower right')
plt.tight_layout()
plt.show()

# ---- Quantitative evaluation ----
print(f"SVR R2 Score: {r2_score(y_svr, svr_model.predict(X_svr)):.4f}")
print(f"SVR RMSE:      {root_mean_squared_error(y_svr, svr_model.predict(X_svr)):.4f}")
print(f"Support Vectors: {len(X_sv)} out of {len(X_svr)} points ({len(X_sv)/len(X_svr)*100:.1f}%)")

```



SVR R² Score: 0.9864
SVR RMSE: 0.1157
Support Vectors: 18 out of 120 points (15.0%)